Blueprint: Herbrand, the unit index, and class-group Stickelberger (flt-cyclotomic-nt)

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Abstract

This blueprint maps `flt-cyclotomic-nt`, the number-theoretic core shared by the engine and the regular-prime route. It develops four exports:

- a kernel-checkable Bernoulli-numerator (irregularity) certificate;
- Herbrand's theorem, in eigenspace form, from the class-group Stickelberger relation (assembled from the single-prime core of `flt-stickelberger`);
- the cyclotomic-unit index $[E : C] = h^+$ (Washington Thm 8.2); and
- the Cauchy–Genocchi Case I criterion.

The single-prime Stickelberger relation is consumed from `flt-stickelberger` and appears here as an external leaf. Nodes link by `\lean` to their declarations (`CyclotomicNT`, `FaithSpike`, `IrrCertNat`, and a few top-level names).

0.1 The kernel Bernoulli / irregularity certificate

Definition 0.1.1 (Pure-`Nat` Bernoulli recurrence). *`bModN` computes the Bernoulli numerator $B_k \bmod p$ by a pure-`Nat` recurrence that the kernel can `decide` directly.*

Theorem 0.1.2 (Faithfulness). *The recurrence is faithful: if `bModN` is nonzero then $p \nmid B_k$, so k is not an irregular index. This is the dual-use check — it discharges the Case I hypothesis at $k = p - 3$ and certifies regularity over all k.*

Definition 0.1.3 (Irregular index). *`IsIrregularIndex` $p\,k$: p divides the numerator of B_k.*

Definition 0.1.4 (Irregularity list certificate). *`irrListCert` $p\,L$: the Boolean certificate that L is exactly the list of irregular indices of p.*

Theorem 0.1.5 (Nat certificate bridge). *A kernel-`decide`d `Nat` certificate implies `irrListCert`, tying the fast check to the semantic irregular-index list.*

0.2 Class-group Stickelberger and Herbrand

Definition 0.2.1 (Single-prime Stickelberger (external)). *The integral Stickelberger relation for a single prime, in principal-ideal form, consumed from `flt-stickelberger` (`stickelbergerProd_prime_isPrincipal_of_descent`).*

Theorem 0.2.2 (Stickelberger annihilates the class group). *The integral Stickelberger element annihilates $\mathrm{Cl}(()\mathcal{O}_K)$, obtained by reducing the all-ideals statement to the single-prime core over the prime factorization.*

Theorem 0.2.3 (All-ideals reduction). *The bridge lemma extending annihilation from principal primes to arbitrary ideals, the form the downstream arguments use.*

Theorem 0.2.4 (Herbrand's theorem). *For odd k in range, $p \mid \mathrm{num}(B_k)$ forces the ω^{1-k} eigencomponent of the p-class group to be nontrivial — the eigenspace projection of Stickelberger's annihilation.*

Theorem 0.2.5 (Herbrand, eigenprojection form). *The explicit eigenspace-projection statement consumed by the Case I and regular-prime arguments.*

0.3 The cyclotomic-unit index

Theorem 0.3.1 (Cyclotomic-unit index, Washington Thm 8.2). *The index of the group C of cyclotomic units in the full unit group E of K^+ equals the plus class number: $[E : C] = h^+$. The proof runs through the analytic class-number formula and a regulator/determinant computation (the `Eigen`, `Dedekind`, and L-value modules).*

Theorem 0.3.2 (Unit index, proof form). *The packaged form of the index theorem exported to the engine, where it lifts the Q_i certificate to the Vandiver condition $p \nmid h^+$.*

0.4 Case I criterion and the prime predicates

Theorem 0.4.1 (Cauchy–Genocchi Case I criterion). *If $p \nmid B_{p-3}$ (equivalently $p - 3$ is not an irregular index), then the first case of Fermat's Last Theorem holds for p. The load-bearing step is the unconditional implication that a Case I factorization with a good Mirimanoff ratio forces $p \mid B_{p-3}$.*

Definition 0.4.2 (Vandiver prime). `IsVandiverPrime` *p: $p \nmid h^+$ — the condition the engine needs, characterized via the unit index and certified per prime by the Q_i test.*

Definition 0.4.3 (Regular prime). `IsRegularPrime` *p: $p \nmid h(K)$, equivalently p divides no Bernoulli numerator $B_2, \ldots, B_{p-3}$ (minus part) and $p \nmid h^+$ (plus part).*